The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science
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MAT 3007 — Optimization

Exercise Sheet 5

Exercise E5.1 ((10+12+15=37pts): Max Flow Problem):
Consider the max flow problem on the graph in figure 1 with the orange node being the source node and the green node being the terminal node (the number on each edge is its capacity, see the lecture slide 13). Do the following based on the lecture slides.

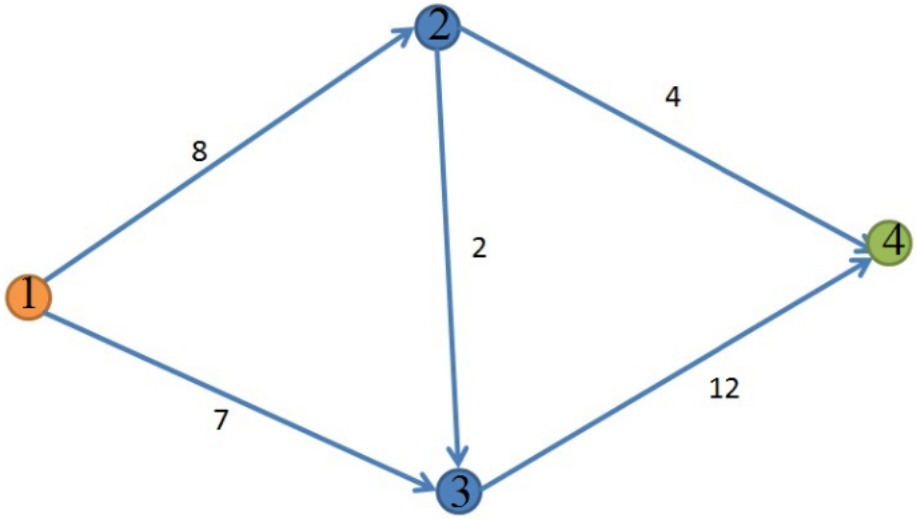


Figure 1: Max flow problem

a) Formulate it as a linear program and solve it using MATLAB.

b) Formulate the dual of this problem and solve an optimal solution via complementarity conditions.

c) Find the corresponding maximum flow and minimum cut of the graph.(Please draw the cut on figure 1).

Exercise E5.2 (Problem 2 (12 + 17 + 17 + 17 = 63pts): Sensitivity Analysis):

Consider the following linear program:

$$\begin{aligned} \max \quad & 5x_1 + 10x_2 \\ \text{s.t.} \quad & x_1 + 3x_2 \leq 50 \\ & 4x_1 + 2x_2 \leq 60 \\ & x_1 \leq 5 \\ & x_1, x_2 \geq 0 \end{aligned}$$

B	0	0	$\frac{10}{3}$	0	$\frac{5}{3}$	175
2	0	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	15
4	0	0	$-\frac{2}{3}$	1	$-\frac{10}{3}$	10
1	1	0	0	0	1	5

Table 1: Simplex Tableau for Sensitivity Analysis

Table 1 gives the final simplex tableau when solving the standard form of the above problem. From the optimal simplex tableau, you are supposed to solve the following questions.

a) What is the optimal solution and the optimal value of the original problem?

b) In what range can we change the coefficient of the first constraint $b_1 = 50$ (the one appearing in the constraint $x_1 + 3x_2 \leq 50$) so that the current optimal basis of standard LP still remains optimal?

c) If we change $b_1 = 50$ to $b_1 = 60$, what will be the new optimal primal solution and the new optimal value?

d) In what range can we change the objective coefficient $c_2 = 10$ so that the current optimal basis of standard LP still remains optimal?

Sheet 5 is due on **Mar, 28th**. Submit your solutions before **Mar, 28th**, **11:00pm**.